Rational 5-torsion and the 2-primary torsion of geometrically split abelian surfaces

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Theorem. *Let $A/\mathbb{Q}$ be an abelian surface that is simple over $\mathbb{Q}$ and geometrically split. If $A(\mathbb{Q})$ has a point of order 5, then*

$$\#A(\mathbb{Q})[2^\infty] \le 4.$$

Corollary. *No genus-2 Jacobian over $\mathbb{Q}$ that is geometrically split and simple over $\mathbb{Q}$ has a rational point of order 5 and at least 8 rational points of 2-power order. In particular [40] and [2, 20] do not occur as the rational torsion subgroup of such a Jacobian. Neither do $[2, 2, 10]$, $[2, 2, 20]$ and $[2, 2, 2, 10]$, which the census proof [P2210] had already excluded.*

Here "geometrically split" means that $A_{\bar{\mathbb{Q}}}$ is isogenous to a product of two elliptic curves. We impose no restriction on the degree of this isogeny, on the field of definition of the geometric endomorphisms or on the integral endomorphism ring, and we assume no polarization. The bound 4 is attained, by Jacobians with rational torsion subgroup $[2, 10]$ (Remark 4). The theorem strengthens the result of [P2210] (a point of order 5 forces rational 2-rank at most 2, for principally polarized surfaces).

Outline. We follow the case distinction of [P2210]. In the CM case, and when the field of definition of the geometric endomorphisms is not quadratic, the arguments of [P2210] need only $40 \mid \#\widetilde{A}(\mathbb{F}_3)$ or no information about the 2-primary part at all (Section 2). In the remaining cases A contains, over a quadratic field K, an elliptic curve E with $E^\sigma \ne E$. In [P2210] a principal polarization and rational 2-rank 3 were used to find an elliptic curve in A_K with full K-rational 2-torsion. Here a curve with full K-rational 2-torsion and a K-point of order 5, K-isogenous to E, comes from $\#A(\mathbb{Q})[2^\infty] \ge 8$ alone. The exact sequence used for $[2, 2, 14]$ [P2214, Proposition 2] puts 2-primary torsion of order at least 4 on E or on A_K/E^σ. By Kenku–Momose no elliptic curve over a quadratic field has a point of order 20, so with the point of order 5 this torsion is the full 2-torsion of a curve in the isogeny class (Lemma 9). From there the analysis of quadratic points on $X_1(2, 10)$ in [P2210] goes through with 8 in place of 40 (Section 4).

1 Conventions and inputs

We use the conventions of [P2210, Section 1]. For a number field k, $G_k = \mathrm{Gal}(\bar{\mathbb{Q}}/k)$. For a quadratic field K, σ is its nontrivial automorphism and $\tilde{\sigma} \in G_{\mathbb{Q}}$ a lift of σ. For an elliptic curve $E \subset A_K$ defined over K, $E^\sigma = \tilde{\sigma}(E) \subset A_K$ is its conjugate; it is defined over K and does not depend on the lift. We use the following results; the labels (I1)–(I4), (I7) and (I9) are those of [P2210, Section 1], and (I10) and (I11) are added here.

- **(I1) Reduction** [P2210, (I1)], from Serre–Tate [ST]. Let K' be a finite extension of $\mathbb{Q}_p$ with residue field k and A an abelian variety over K'. (a) Good reduction is

preserved by K'-isogenies. (b) If A has good reduction $\widetilde{A}$ and m is prime to p, then $A(K')[m]$ injects into $\widetilde{A}(k)$. (c) If A has potentially good reduction, then it has good reduction over a finite extension of K' with residue field k. (d) An abelian surface B over $\mathbb{F}_q$ has $\#B(\mathbb{F}_q) \le (1+\sqrt{q})^4$.

- **(I2) The case distinction** [P2210, (I2)]. A geometrically split abelian surface $A/\mathbb{Q}$ is of exactly one of the types **(N)** ($A_{\bar{\mathbb{Q}}}$ is isogenous to $E_1 \times E_2$ with E_1, E_2 not isogenous), **(CM)** ($A_{\bar{\mathbb{Q}}}$ is isogenous to E^2 with E a CM curve) and **(S)** ($A_{\bar{\mathbb{Q}}}$ is isogenous to E^2 with E without CM). In case (S), L is the smallest field over which all geometric endomorphisms are defined, $\Gamma = \mathrm{Gal}(L/\mathbb{Q})$, and $\Gamma \neq 1$ if A is simple over $\mathbb{Q}$.

- **(I3)** [P63, Lemma 2]. Let X be an abelian surface over a number field k, $C \subset X$ an elliptic curve defined over k, and $Q \in X(k)$ of prime order q. Then either $Q \in C(k)$, or the image of Q in $(X/C)(k)$ has order q.

- **(I4)** [P63, Lemma 3]. In case (S), every elliptic curve in $A_{\bar{\mathbb{Q}}}$ is defined over L.

- **(I7) Weil restriction** [P2210, (I7)]. For an elliptic curve E over a quadratic field K, $(\mathrm{Res}_{K/\mathbb{Q}} E)(\mathbb{Q}) = E(K)$ and $(\mathrm{Res}_{K/\mathbb{Q}} E)_K \cong E \times E^\sigma$. If E is the base change of an elliptic curve $E_0/\mathbb{Q}$, then $\mathrm{Res}_{K/\mathbb{Q}} E$ is isogenous over $\mathbb{Q}$ to $E_0 \times E_0^{(K)}$.

- **(I9) Kubert's normal form** [P2210, (I9)], from [Ku, Table 3]. For t in a field k of characteristic 0 put

$$c = -\frac{t(t-1)(2t-1)}{t^2-3t+1}, \qquad b = \frac{t^3(t-1)(2t-1)}{(t^2-3t+1)^2}, \qquad E_t:\ y^2+(1-c)xy-by = x^3-bx^2.$$

  If E_t is an elliptic curve, $(0,0)$ has order 10, and every pair (E', P) of an elliptic curve over k and a point $P \in E'(k)$ of order 10 is isomorphic over k to $(E_t, (0,0))$ for some such $t \in k$.

- **(I10) Torsion over quadratic fields** [KM, Theorem (0.1)], in the form quoted in [P2214, (I1)]. Let N be a product of powers of $2, 3, 5, 7, 11, 13$ and $m \mid N$. If $X_1(m,N)$ has positive genus and (m,N) is not one of $(1,11)$, $(1,13)$, $(1,14)$, $(1,15)$, $(1,16)$, $(1,18)$, $(2,10)$, $(2,12)$, then $Y_1(m,N)(k) = \emptyset$ for every quadratic field k. The curve $X_1(20)$ has genus 3 (certificate, part K). Hence *no elliptic curve over a quadratic field has a point of order* 20.

- **(I11)** [P63, Section 3], [P2210, Section 5]. If A is of type (N) and simple over $\mathbb{Q}$, the two elliptic curves of $A_{\bar{\mathbb{Q}}}$ are defined over a quadratic field K and exchanged by σ. An elliptic curve $E \subset A_K$ with $E^\sigma = E$ is stable under $G_{\mathbb{Q}}$, hence an elliptic curve in A defined over $\mathbb{Q}$ [P2214, Section 2].

From [P2210] we use the following results, with its numbering.

- **Lemma 1.** No abelian surface over $\mathbb{F}_3$ has exactly 40 points.

- **Proposition 2.** If A is of type (S), simple over $\mathbb{Q}$, and $A(\mathbb{Q})[5] \neq 0$, then Γ is cyclic of order 2 or 4.

- **Proposition A.** If $A_{\bar{\mathbb{Q}}}$ is isogenous to the square of an elliptic curve without CM and the field of definition of the geometric endomorphisms of A is a cyclic quartic extension of $\mathbb{Q}$, then $\#A(\mathbb{Q})[2^\infty] \le 4$.

Neither Lemma 1, nor Proposition 2, nor Proposition A uses a principal polarization [P2210, Remark 3]. Proposition 2 rests on the residual-representation lemmas of [P63, Section 6], which use a polarization of A over $\mathbb{Q}$ of arbitrary degree; every abelian variety over $\mathbb{Q}$ has one.

The following results of the quadratic case of [P2210, Section 5] concern an abelian surface $A/\mathbb{Q}$ that is simple over $\mathbb{Q}$, a quadratic field K, and an elliptic curve $E \subset A_K$ defined over K with $E^\sigma \neq E$. In [P2210] they are stated under the standing assumption of that section. The proof of Lemma 5 uses only these hypotheses [P2210, Remark 2], and Lemmas 6, 7 and Proposition 8 concern elliptic curves over K only.

- **Lemma 5 (Weil restriction).** (a) A is isogenous over $\mathbb{Q}$ to $\mathrm{Res}_{K/\mathbb{Q}} E'$ for every elliptic curve E' over K that is K-isogenous to E. (b) If p is an odd prime, inert in K, at which A has good reduction, then every such E' has good reduction at pO_K, and $\#\widetilde{A}(\mathbb{F}_p) = \#\widetilde{E'}(O_K/pO_K)$.

- **Lemma 6 (a Chebotarev criterion).** Let $K = \mathbb{Q}(\sqrt{d})$ and $E : y^2 = (x-e_1)(x-e_2)(x-e_3)$ with $e_i \in K$ distinct, $N_{ij} = N_{K/\mathbb{Q}}(e_i - e_j)$. If $8 \mid \#\widetilde{E}(O_K/pO_K)$ for all but finitely many primes p inert in K, then for a suitable numbering $N_{ij} \in \mathbb{Q}^{\times 2}$, and $N_{ik} \in \mathbb{Q}^{\times 2}$ or $N_{jk} \in \mathbb{Q}^{\times 2}$ or $dN_{ik}N_{jk} \in \mathbb{Q}^{\times 2}$.

- **Lemma 7 ($\mathbb{Z}/2 \times \mathbb{Z}/10$ over quadratic fields).** Let $\mathcal{X}$ be the elliptic curve $y^2 = x^3 - 2x^2 + 1$ over $\mathbb{Q}$, with origin the point at infinity O (Cremona 20a2), let E_t be as in (I9), and put

$$e_1 = -\frac{t^3(t-1)}{t^2-3t+1}, \qquad B_t = \frac{-8t^4+12t^3-4t+1}{(t^2-3t+1)^2},$$

$$\rho = \frac{(2t-1)^2 s}{(t^2-3t+1)^2}, \qquad e_{2,3} = \frac{-B_t \pm \rho}{8}.$$

  (a) $\mathcal{X}$ has rank 0 and $\mathcal{X}(\mathbb{Q}) = \{O, (0,\pm 1), (1,0), (2,\pm 1)\}$. (b) Let K be a quadratic field and E' an elliptic curve over K with $E'(K) \supseteq \mathbb{Z}/2 \times \mathbb{Z}/10$. Then $E' \cong E_t$ over K for some $t \in K$ such that $s^2 = (2t-1)(4t^2-2t-1)$ for some $s \in K$, and E' is isomorphic over K to $y^2 = (x-e_1)(x-e_2)(x-e_3)$. The point $Q = (2t, s) \in \mathcal{X}(K)$ does not lie in $\mathcal{X}(\mathbb{Q})$. (c) Put $R = Q + Q^\sigma \in \mathcal{X}(\mathbb{Q})$. If $R = O$, then $t \in \mathbb{Q}$. If $R \neq O$, then the line through Q and Q^σ passes through $-R$ and has a slope $m \in \mathbb{Q}$; the x-coordinates of Q and Q^σ are the roots of the monic quadratic

$$q_{R,m}(X) = \frac{X^3 - 2X^2 + 1 - \big(y(-R) + m(X - x(-R))\big)^2}{X - x(-R)} \in \mathbb{Q}[X],$$

  which is irreducible over $\mathbb{Q}$, and $K = \mathbb{Q}\big(\sqrt{\operatorname{disc} q_{R,m}}\big)$.

- **Proposition 8.** Let $R \in \mathcal{X}(\mathbb{Q}) \setminus \{O\}$ and $m \in \mathbb{Q}$ be such that $q_{R,m}$ is irreducible over $\mathbb{Q}$. Let x_Q be a root of $q_{R,m}$, $K = \mathbb{Q}(x_Q)$, $t = x_Q/2$ and $s = y(-R) + m(x_Q - x(-R))$, so that $Q = (x_Q, s) \in \mathcal{X}(K)$, and suppose that E_t is an elliptic curve. Then the conclusion of Lemma 6 fails for $y^2 = (x-e_1)(x-e_2)(x-e_3)$ over K, with e_1, e_2, e_3 as above.

The computational inputs of Lemma 1, Lemma 7 and Proposition 8 are certified by `qsimple_2210_checks.m` of [P2210].

2 The CM case and endomorphism fields that are not quadratic

Type (CM). Let A be of type (CM) with a point $P_5 \in A(\mathbb{Q})$ of order 5, and suppose that $\#A(\mathbb{Q})[2^\infty] = 2^k \geq 8$. As in [P2210, Section 2], A has potentially good reduction at 3, and by (I1)(c) good reduction $\widetilde{A}$ over a finite extension of $\mathbb{Q}_3$ with residue field $\mathbb{F}_3$. The subgroup of $A(\mathbb{Q})$ generated by P_5 and $A(\mathbb{Q})[2^\infty]$ has order $5 \cdot 2^k$, prime to 3, so by (I1)(b) $5 \cdot 2^k$ divides $\#\widetilde{A}(\mathbb{F}_3)$, which is at most $(1+\sqrt{3})^4 < 56$ by (I1)(d). Since $5 \cdot 16 = 80 > 56$, this forces $2^k = 8$ and $\#\widetilde{A}(\mathbb{F}_3) = 40$, contradicting Lemma 1. So $\#A(\mathbb{Q})[2^\infty] \leq 4$. This case uses neither simplicity over $\mathbb{Q}$ nor a polarization.

Type (S). Let A be of type (S), simple over $\mathbb{Q}$, with a point of order 5. By Proposition 2, $\Gamma \cong C_2$ or $\Gamma \cong C_4$. If $\Gamma \cong C_4$, then $\#A(\mathbb{Q})[2^\infty] \leq 4$ by Proposition A.

It remains to treat type (S) with $\Gamma \cong C_2$ and type (N).

3 Moving 2-primary torsion to an elliptic curve

Lemma 9. Let $A/\mathbb{Q}$ be an abelian surface, K a quadratic field and $E \subset A_K$ an elliptic curve defined over K with $E^\sigma \neq E$. Put $Y = A_K/E^\sigma$, and suppose that $A(\mathbb{Q})$ has a point of order 5. Then:

(a) $E(K)[2^\infty]$ and $Y(K)[2^\infty]$ are killed by 2;

(b) $A(\mathbb{Q})[2^\infty]$ is killed by 4 and has order at most 16; in particular $A(\mathbb{Q})$ has no point of order 8;

(c) if $\#A(\mathbb{Q})[2^\infty] \geq 8$, there is an elliptic curve E' over K, K-isogenous to E, such that $E'(K) \supseteq \mathbb{Z}/2 \times \mathbb{Z}/10$.

Proof. Step 1: an exact sequence (as in [P2214, Proposition 2]). The curves E and E^σ are distinct elliptic curves in A_K, so $E \cap E^\sigma$ is finite. Let $\pi : A_K \to Y$ be the quotient map, defined over K. Its restriction to E has the finite kernel $E \cap E^\sigma$, so it is a K-isogeny $E \to Y$. If $P \in A(\mathbb{Q})$ and $\pi(P) = 0$, then $P \in E^\sigma$, hence $P = \tilde{\sigma}(P) \in \tilde{\sigma}(E^\sigma) = \tilde{\sigma}^2(E) = E$, as $\tilde{\sigma}^2 \in G_K$; so $P \in E(\bar{\mathbb{Q}}) \cap A(K) = E(K)$. Therefore

$$0 \to A(\mathbb{Q})[2^\infty] \cap E(K) \to A(\mathbb{Q})[2^\infty] \xrightarrow{\pi} Y(K)[2^\infty]$$

is exact, and $\#A(\mathbb{Q})[2^\infty] \leq \#E(K)[2^\infty] \cdot \#Y(K)[2^\infty]$.

Step 2: a point of order 5. Let $P_5 \in A(\mathbb{Q})$ have order 5. By (I3) over K, applied to $E^\sigma \subset A_K$, either $P_5 \in E^\sigma$, and then $P_5 \in E(K)$ as in Step 1, or $\pi(P_5) \in Y(K)$ has order 5. So for some $C' \in \{E, Y\}$ there is a point $P \in C'(K)$ of order 5.

Step 3: combining (as in [P63, Lemma 1], with a group in place of a point). Let $C \in \{E, Y\}$ and $T = C(K)[2^\infty]$. If $C' = C$, put $E' = C$, let a be the identity of C, and put $T' = T$ and $P' = P$. Otherwise let $f : C \to C'$ be the K-isogeny $\pi|_E$ or its dual. Let H be the 5-primary part of $\ker f$; it is a characteristic subgroup of the G_K-stable group $\ker f$, hence G_K-stable. Put $E' = C/H$, let $a : C \to E'$ be the quotient map and write $f = b \circ a$ with $b : E' \to C'$ defined over K. The degree of a is a power of 5, so a is injective on $C[2^\infty]$ and $a(T) \subseteq E'(K)$ is isomorphic to T. The degree of b is prime to 5; since $b \circ \hat{b} = [\deg b]$ is injective on $C'[5]$, so is the dual isogeny $\hat{b} : C' \to E'$, which is defined over K, and $\hat{b}(P) \in E'(K)$ has order 5; put $T' = a(T)$ and $P' = \hat{b}(P)$. In both cases E' is K-isogenous to E, and

$$E'(K) \supseteq T' \oplus \langle P' \rangle, \qquad T' \cong T, \quad P' \text{ of order } 5.$$

(a) If T had an element of order 4, then $E'(K)$ would have a point of order 20, contradicting (I10). So T is killed by 2, for $C = E$ and for $C = Y$.

(b) By (a) and Step 1, $\#A(\mathbb{Q})[2^\infty] \le 4 \cdot 4$. For $x \in A(\mathbb{Q})[2^\infty]$ we have $2\pi(x) = 0$ by (a), so $2x \in A(\mathbb{Q})[2^\infty] \cap \ker \pi \subseteq E(K)[2^\infty]$, and $4x = 0$ by (a).

(c) If $\#A(\mathbb{Q})[2^\infty] \ge 8$, Step 1 and (a) give $C \in \{E, Y\}$ with $\#T \ge 4$ and T killed by 2, so $T = C[2]$. With E' and a as in Step 3, $T' = a(T)$ is a subgroup of $E'[2]$ of order 4, so $T' = E'[2]$. Hence $E'[2] \subseteq E'(K)$ and $E'(K) \supseteq \mathbb{Z}/2 \times \mathbb{Z}/2 \times \mathbb{Z}/5 \cong \mathbb{Z}/2 \times \mathbb{Z}/10$. □

Part (b) already excludes rational 2-primary subgroups containing $\mathbb{Z}/8$, without any Frobenius argument; in particular it excludes [40] in the cases of this section. The 2-primary groups $\mathbb{Z}/2 \times \mathbb{Z}/4$, $(\mathbb{Z}/2)^3$, $(\mathbb{Z}/4)^2$, $(\mathbb{Z}/2)^2 \times \mathbb{Z}/4$ and $(\mathbb{Z}/2)^4$ need part (c) and Section 4.

4 The quadratic case

Theorem Q. Let $A/\mathbb{Q}$ be an abelian surface that is simple over $\mathbb{Q}$. Suppose that there are a quadratic field K and an elliptic curve $E \subset A_K$ defined over K with $E^\sigma \ne E$. If $A(\mathbb{Q})$ has a point of order 5, then $\#A(\mathbb{Q})[2^\infty] \le 4$.

Proof. Suppose that $\#A(\mathbb{Q})[2^\infty] \ge 8$. By Lemma 9(c) there is an elliptic curve E' over K, K-isogenous to E, with $E'(K) \supseteq \mathbb{Z}/2 \times \mathbb{Z}/10$. By Lemma 5(a), A is isogenous over $\mathbb{Q}$ to $\mathrm{Res}_{K/\mathbb{Q}} E'$. Let p be an odd prime, inert in K, at which A has good reduction. By (I1)(b), $A(\mathbb{Q})[2^\infty]$ injects into $\widetilde{A}(\mathbb{F}_p)$, so $8 \mid \#\widetilde{A}(\mathbb{F}_p) = \#\widetilde{E'}(O_K/pO_K)$ by Lemma 5(b). This holds for all but finitely many primes p inert in K. Let $E' \cong E_t$ and $Q \in \mathcal{X}(K)$ be as in Lemma 7.

- If $Q + Q^\sigma = O$, then $t \in \mathbb{Q}$, and E' is the base change of the elliptic curve E_t over $\mathbb{Q}$. By (I7), $\mathrm{Res}_{K/\mathbb{Q}} E'$ is isogenous over $\mathbb{Q}$ to $E_t \times E_t^{(K)}$, so A is not simple over $\mathbb{Q}$. This is a contradiction.

- Otherwise $R = Q + Q^\sigma \ne O$. By Lemma 7(c), $q_{R,m}$ is irreducible over $\mathbb{Q}$, with roots $x(Q)$ and $x(Q^\sigma)$, and $K = \mathbb{Q}(x(Q))$ because $x(Q) \notin \mathbb{Q}$. (If $x(Q) \in \mathbb{Q}$, then $Q^\sigma \ne Q$ has the same x-coordinate, so $Q^\sigma = -Q$ and $R = O$.) As $Q = (2t, s)$ lies on the line through $-R$ with slope m, we have $t = x(Q)/2$ and $s = y(-R) + m(x(Q) - x(-R))$, and $E_t \cong E'$ is an elliptic curve. So the hypotheses of Proposition 8 hold with $x_Q = x(Q)$. The curve E' is isomorphic over K to $y^2 = (x - e_1)(x - e_2)(x - e_3)$, so the two curves have isomorphic reductions at all but finitely many primes of K, and Lemma 6 applies to $y^2 = (x - e_1)(x - e_2)(x - e_3)$. Its conclusion contradicts Proposition 8.

Hence $\#A(\mathbb{Q})[2^\infty] \le 4$. □

Type (S) with $\Gamma \cong C_2$, and type (N). Let A be simple over $\mathbb{Q}$. If A is of type (S) with $\Gamma \cong C_2$, put $K = L$; by (I4) every elliptic curve $E \subset A_{\bar{\mathbb{Q}}}$ is defined over K. If A is of type (N), let K be the quadratic field of (I11) and E one of the two elliptic curves of $A_{\bar{\mathbb{Q}}}$. In both cases $E^\sigma \ne E$ by (I11), since A is simple over $\mathbb{Q}$. So Theorem Q applies.

5 Proof of the Theorem and consequences for the census

Proof of the Theorem. Let $A/\mathbb{Q}$ be simple over $\mathbb{Q}$ and geometrically split, with a rational point of order 5. By (I2), A is of type (N), (CM) or (S). In type (CM), $\#A(\mathbb{Q})[2^\infty] \le 4$ by Section 2. In type (S), $\Gamma \cong C_2$ or C_4 by Proposition 2; for $\Gamma \cong C_4$ the bound holds by Proposition A, and for $\Gamma \cong C_2$ by Theorem Q. In type (N) the bound holds by Theorem Q. □

Proof of the Corollary. The Jacobian of a curve of genus 2 over $\mathbb{Q}$ is an abelian surface. The groups [40] and [2, 20] have a point of order 5 and 2-primary parts $\mathbb{Z}/8$ and $\mathbb{Z}/2 \times \mathbb{Z}/4$

of order 8; the groups $[2,2,10]$, $[2,2,20]$ and $[2,2,2,10]$ have 2-primary parts of orders 8, 16 and 16. □

On the census, [40] and [2, 20] become impossible in the class "geometrically split, simple over $\mathbb{Q}$". Both have examples split over $\mathbb{Q}$ (the census curves 40.b and 2.20.b), so their combined geometrically split cells keep their grades. For $[2,2,10]$, $[2,2,20]$ and $[2,2,2,10]$ the census entries already cite [P2210]; the theorem removes its principal polarization hypothesis. In the class "geometrically split, simple over $\mathbb{Q}$", the group [20] (2-primary part $\mathbb{Z}/4$) is not decided by this argument.

6 Remarks

1. *Where the hypotheses are used.* Simplicity over $\mathbb{Q}$ gives $E^\sigma \neq E$ in the quadratic case, excludes the base-change alternative in Theorem Q, and enters Proposition 2. It is needed: the census lists examples split over $\mathbb{Q}$ of [40], [2, 20] and $[2,2,10]$. The point of order 5 is also needed: the census curves 2.2.8.c and 2.2.2.4.c are geometrically split and simple over $\mathbb{Q}$, with rational torsion subgroups $[2,2,8]$ and $[2,2,2,4]$, whose 2-primary parts have order 32. In the quadratic case the rational point of order 5 of A is used only in Step 2 of Lemma 9, to put a K-point of order 5 on E or Y.

2. *The polarization in the earlier theorem.* By [P2210, Remark 3], the principal polarization was used there only in Lemma 4. Rational 2-rank at least 3 gives $\#A(\mathbb{Q})[2^\infty] \geq 8$. So in the last paragraph of [P2210, Section 5], Lemma 9(c) can replace Lemma 4 and Lemma 5(c) together, with Lemma 5(a) applied to the original curve E (the curve E' of Lemma 9 need not lie in A_K); the argument then becomes the proof of Theorem Q. Hence the theorem of [P2210] holds for every abelian surface over $\mathbb{Q}$ that is simple over $\mathbb{Q}$ and geometrically split; it is also a special case of the theorem above.

3. *Isogenies and $\mathbb{Q}$-curves.* Rational 2-primary torsion is not an isogeny invariant, and $\mathbb{Q}$-isogenies of a Weil restriction $\mathrm{Res}_{K/\mathbb{Q}} E'$ of a $\mathbb{Q}$-curve, for instance $(2,2)$-isogenies, can enlarge it. Lemma 9 bounds the 2-primary torsion of every surface in the $\mathbb{Q}$-isogeny class by that of two curves in the K-isogeny class of E. If $E'(K) \supseteq \mathbb{Z}/2 \times \mathbb{Z}/10$ and E' is not a base change, Proposition 8 and Lemma 6 give infinitely many inert p with $8 \nmid \#\widetilde{E'}(\mathbb{F}_{p^2}) = \#\widetilde{A}(\mathbb{F}_p)$, for every A isogenous to $\mathrm{Res}_{K/\mathbb{Q}} E'$. For example, the two elliptic curves over quadratic fields in the LMFDB with torsion $\mathbb{Z}/2\times\mathbb{Z}/10$ that are not base changes, 2.2.13.1-36.1-a3 and 2.2.17.1-36.1-a4, have $\#\widetilde{E}(\mathbb{F}_{121}) = 100$ (certificate, part S).

4. *Sharpness.* Let E be the curve 2.2.13.1-36.1-a3, $y^2 + (1+w)xy + (1+w)y = x^3 + (-115+48w)x + (218-96w)$ over $K = \mathbb{Q}(w)$, $w^2 - w - 3 = 0$, and $A = \mathrm{Res}_{K/\mathbb{Q}} E$. The certificate (part S) checks that $E(K)_{\mathrm{tors}} \cong \mathbb{Z}/2\times\mathbb{Z}/10$, that $j(E) = 476379541/236196$ is not integral (so E has no CM), that $E \cong E^\sigma$ over K, and that the Frobenius polynomial $T^4 - 10T^2 + 49$ of A at the inert prime 7 is irreducible. Hence A is simple over $\mathbb{Q}$. As $\mathrm{End}(E_{\bar{\mathbb{Q}}}) = \mathbb{Z}$ and $E \cong E^\sigma$ over K, every geometric endomorphism of A is defined over K, so $L \subseteq K$, and $L \neq \mathbb{Q}$ by (I2); thus A is of type (S) with $\Gamma \cong C_2$. It is principally polarized (the product polarization of $E \times E^\sigma$ descends), and $A(\mathbb{Q})_{\mathrm{tors}} = E(K)_{\mathrm{tors}} \cong \mathbb{Z}/2\times\mathbb{Z}/10$ by (I7). This A is not a Jacobian: as $\mathrm{End}(E_{\bar{\mathbb{Q}}}) = \mathbb{Z}$, the principal polarizations of $A_{\bar{\mathbb{Q}}} \cong E^2$ correspond to positive definite unimodular binary quadratic forms over $\mathbb{Z}$, each equivalent to $x^2 + y^2$, the product polarization.

   The bound is also attained by a Jacobian. Gluing E and E^σ along $\psi(\alpha_i, 0) = (\sigma(\alpha_i), 0)$, where $\alpha_1, \alpha_2, \alpha_3 \in K$ are the x-coordinates of the points of order 2 of E [HLP, Propo-

sition 4], gives a curve over K that descends to the genus-2 curve

$$C:\ y^2 = 63x^6 + 6x^5 + 184x^4 + 184x^2 - 6x + 63$$

over $\mathbb{Q}$, the curve 6084.d1 of the LMFDB table `g2c_curves_new`. The certificate (part J) checks that $J(\mathbb{Q})_{\mathrm{tors}} \cong \mathbb{Z}/2 \times \mathbb{Z}/10$ for $J = \mathrm{Jac}(C)$; that the L-polynomial $49T^4 - 10T^2 + 1$ of C at 7 is irreducible, so J is simple over $\mathbb{Q}$; and that over K, C is isomorphic to the curve $y^2 = h(X)$ of [HLP, Proposition 4] with h even, whose two elliptic quotients $(X, y) \mapsto (X^2, y)$ and $(X, y) \mapsto (1/X^2, y/X^3)$ are K-isomorphic to E and to $E^\sigma \cong E$; it also checks the non-degeneracy condition of [HLP, Proposition 4]. The two quotient maps are defined over K, so J_K is K-isogenous to E^2 and, as $\mathrm{End}(E_{\bar{\mathbb{Q}}}) = \mathbb{Z}$, every geometric endomorphism of J is defined over K. Since J is simple over $\mathbb{Q}$, it is of type (S) with $\Gamma \cong C_2$. The census lists [2, 10] in the class "geometrically split, simple over $\mathbb{Q}$" through 2.10.b (LMFDB 256.a.512.1), whose endomorphism field is cyclic quartic. In the LMFDB table `g2c_curves_new`, five curves with torsion [2, 10] are simple over $\mathbb{Q}$ and geometrically split: 6084.d1, with Sato–Tate group E_2, and 256.a1 (that is, 256.a.512.1), 16900.i1, 28900.u1 and 57600.gw1, with Sato–Tate group E_4; in the table `g2c_curves` only 256.a.512.1 occurs.

5. *Validation (not used in the proof).* Every curve in the class "geometrically split, simple over $\mathbb{Q}$" with a rational point of order 5, in the census and in both LMFDB genus-2 tables, has 2-primary torsion of order at most 4. Among the 2795 LMFDB elliptic curves E over quadratic fields with $5 \mid \#E(K)_{\mathrm{tors}}$, each of the 2108 for which $\mathrm{Res}_{K/\mathbb{Q}} E$ is certified simple over $\mathbb{Q}$ has a good odd prime $p \le 23$ with $8 \nmid \#\widetilde{\mathrm{Res}_{K/\mathbb{Q}} E}(\mathbb{F}_p)$. Across the 2403 isogeny classes of LMFDB elliptic curves over quadratic fields that contain a curve with a point of order 5, no curve has a point of order 4, as (I10) predicts.

6. *Two derivations.* The argument of Sections 3 and 4 was found independently in two separate analyses, of type (N) and of type (S) with $\Gamma \cong C_2$; each checked the other.

7 Computations

All scripts run in Magma V2.30-1 and consist of assertions.

- `qsimple_40_checks.m`, which runs in under a second. Its log `qsimple_40_checks.log` ends with `ALL_T5SQ_CHECKS_PASSED`. Part K: $X_1(20)$ has genus 3 and $(1, 20)$ is not among the exceptions of (I10). Parts S and J: the facts of Remarks 3 and 4. The SHA-256 hash of the script is `2a06ab43a45d3617f00cc3ffc82b5e88e5ebaf7c255e5805 1c8bd966ace1fce2`, and that of the log is `a7ff82365c9fcddcf8b598df50a23a56 abf8afa71192bcedc1778c1a9dc43535`.

- The published certificate `qsimple_2210_checks.m` of [P2210] certifies Lemma 1, Lemma 7 and Proposition 8. Rerun unchanged, it gives a log byte-identical to the published one.

References

[HLP] E. W. Howe, F. Leprévost and B. Poonen, *Large torsion subgroups of split Jacobians of curves of genus two or three*, Forum Math. **12** (2000), 315–364, Proposition 4.

[KM] M. A. Kenku and F. Momose, *Torsion points on elliptic curves defined over quadratic fields*, Nagoya Math. J. **109** (1988), 125–149, Theorem (0.1). doi:10.1017/S0027763000002816.

[Ku] D. S. Kubert, *Universal bounds on the torsion of elliptic curves*, Proc. London Math. Soc. (3) **33** (1976), 193–237, Table 3. doi:10.1112/plms/s3-33.2.193.

[P63] Census proof of 21 October 2026, *An obstruction to rational* 63*-torsion on geometrically split abelian surfaces*, https://genus-2-torsion.github.io/proofs/qsimple-63-obstruction.html: Lemmas 1–3 and Sections 3 and 6.

[P2210] Census proof of 11 November 2026, *Rational* 5*-torsion and the* 2*-rank of geometrically split Jacobians*, https://genus-2-torsion.github.io/proofs/qsimple-2210-obstruction.html: Sections 1, 2 and 5, Lemma 1, Propositions 2, A and 8, Lemmas 5–7, Remarks 2 and 3.

[P2214] Census proof of 10 November 2026, *Rational* 7*-torsion and the* 2*-rank of geometrically split abelian surfaces*, https://genus-2-torsion.github.io/proofs/qsimple-2214-obstruction.html: (I1), Section 2 and Proposition 2.

[ST] J.-P. Serre and J. Tate, *Good reduction of abelian varieties*, Ann. of Math. (2) **88** (1968), 492–517. doi:10.2307/1970722.